

Assignment 8 (Pipeline system)

- 1) Briefly describe various stages of the MIPS five-stage pipeline system.
2. Single cycle MIPS system is pipelined by subdividing its *data path* into *five pipeline stages*. Briefly describe the **pipeline stages** of the processor.
3. The pipelined MIPS processor is running the following program. Which registers are written and *read* on the **sixth cycle** (where S1, S2, S3, S4, T0, T1, T2, T3, and T4 are 32-bit general-purpose registers)?

```
addi S1, S2, 5
sub  T0, T1, T2
lw   T3, 15(S1)
sw   T4, 72(T0)
Or   T2, S3, S4
```

4. Using a **pipeline sequence diagram**, show the *forwarding* and *stalls* needed to execute the following instructions on the pipelined MIPS system (where S1, S2, S3, S4, S5, S6, S7, S8 and S9 are 32-bit general purpose registers).

```
lw   S6, 10(S5)
and  S3, S6, S7
or   S2, S6, S9
add  S1, S2, S3
sub  S4, S1, S5
xor  S10, S3, S1
```

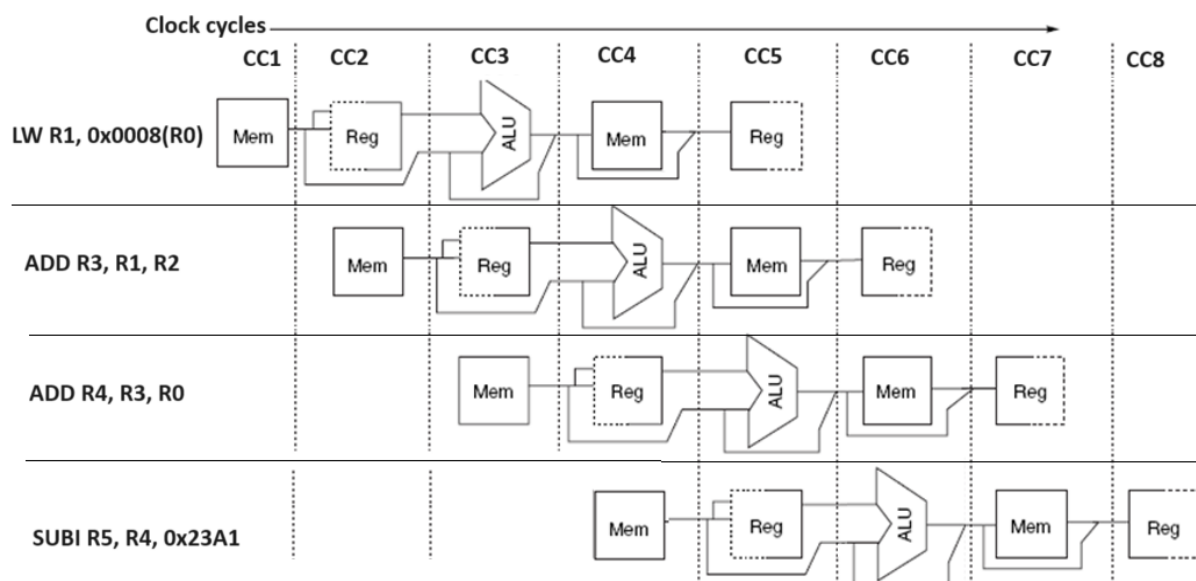
5. Using a **pipeline sequence diagram**, show the *forwarding* and *stalls* needed to execute the following instructions on the pipelined MIPS system (where S0, S1, S2, S3, S4, S5, S6, S7, S8 and S9 are 32-bit general purpose registers).

```
addi S1, S0, 5
add  S4, S1, S5
xor  S10, S3, S4
lw   S6, 8(S5)
or   S3, S6, S7
and  S2, S6, S9
```

6. Assume that the MIPS pipelined system has imposed a stall of **0.78** clock cycles per instruction for a particular user process. Find out the **CPI** and **speed up** of the system.
7. Describe the term **structural hazard** in a pipeline system and how the MIPS system solves the problem with **register file** read and write operation in **1 clock cycle** during its pipeline instruction execution.

8. Consider an **un-pipelined** machine with five stages (Instruction Fetch, Instruction Decode/Register Fetch, Execute/Address Calculation, Memory Access and Write Back). Assume that it has **1ns** clock cycles. The machine uses four cycles for ALU operations and branches and five cycles for memory operations. Assume that the relative frequencies of these operations are 45%, 15%, and 40%, respectively. **Pipelining the machine** adds **1.2 ns** of overhead to the clock. Find out how much **speedup** we will gain in the instruction execution rate. You can ignore any latency impact.

9. What is the *structural hazard*? Consider the pipeline data path of the MIPS system shown in **Figure below**. Is there any structural hazard in the system? If yes, describe the possible ways (if any) to eliminate it from the system.



10. What is the *structural hazard*? Consider the **pipeline data path** of the MIPS system shown in **Figure below**. Is there any structural hazard in the system? If yes, describe the possible ways (if any) to eliminate it.

